Mini Project - E-Commerce Sales Analytics (PowerBI & Tableau)

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Category Revenue by Region (Stacked Bar Chart)
revenue_df = pd.read_csv("category_revenue.csv")
revenue_df = revenue_df.set_index('Region')

fig, axes = plt.subplots(3, 2, figsize=(15, 18))
plt.subplots_adjust(hspace=0.4)

revenue_df.plot(kind='bar', stacked=True, ax=axes[0, 0],
                colormap='Paired', edgecolor='white')
axes[0, 0].set_title('Category Revenue by Region')
axes[0, 0].set_ylabel('Revenue (INR Lakhs)')
axes[0, 0].tick_params(axis='x', rotation=0)
axes[0, 0].set_ylim(0, revenue_df.sum(axis=1).max() * 1.25)
axes[0, 0].legend(title='Category', fontsize=8, ncol=3, loc='upper center')

# 2. Monthly Orders and Returns (Area + Line Chart)
monthly = pd.read_csv("monthly_orders.csv")
axes[0, 1].fill_between(monthly['Month'], monthly['Orders'],
                        color='mediumpurple', alpha=0.35)
axes[0, 1].plot(monthly['Month'], monthly['Orders'], marker='D',
                color='indigo', label='Orders')
axes[0, 1].plot(monthly['Month'], monthly['Returns'], marker='x',
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            color='tomato', label='Returns')
axes[0, 1].set_title('Monthly Orders and Returns')
axes[0, 1].set_xlabel('Month')
axes[0, 1].set_ylabel('Count')
axes[0, 1].legend()

# 3. Order Value Distribution by Payment Method (Box Plot)
orders = pd.read_csv("orders.csv")
sns.boxplot(x='Payment Method', y='Order Value (INR)', data=orders,
        hue='Payment Method', legend=False, palette='Set3',
        showfliers=False, ax=axes[1, 0])
axes[1, 0].set_title('Order Value Distribution by Payment Method')
axes[1, 0].tick_params(axis='x', rotation=15)

# 4. Website Traffic by Day and Time Slot (Heatmap)
traffic_df = pd.read_csv("web_traffic.csv")
day_order = ['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', 'Sun']
pivot = traffic_df.pivot_table(index='Day', columns='Time Slot',
                        values='Sessions', aggfunc='mean')
pivot = pivot.reindex(day_order)
sns.heatmap(pivot, annot=True, fmt=".0f", cmap="YlOrRd", ax=axes[1, 1])
axes[1, 1].set_title('Average Website Sessions by Day and Time Slot')

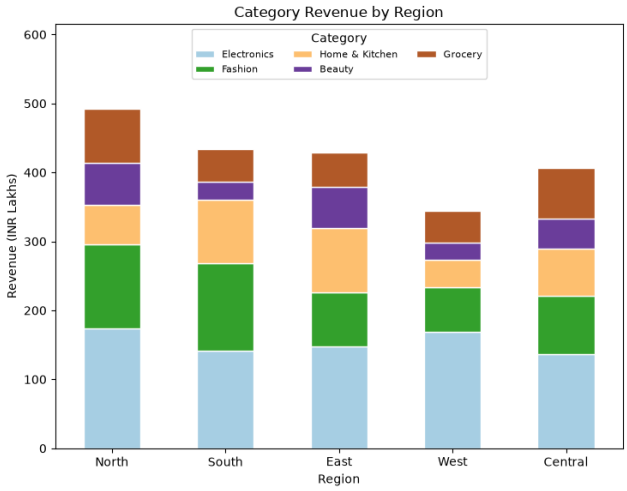
# 5. Order Share by Sales Channel (Donut Chart)
channel_counts = orders['Sales Channel'].value_counts()
axes[2, 0].pie(channel_counts, labels=channel_counts.index,
            autopct='%1.1f%%', startangle=90, pctdistance=0.78,
            colors=['#4c72b0', '#55a868', '#c44e52', '#dd8452'],
            wedgeprops={'width': 0.45, 'edgecolor': 'white'})
axes[2, 0].set_title('Order Share by Sales Channel')

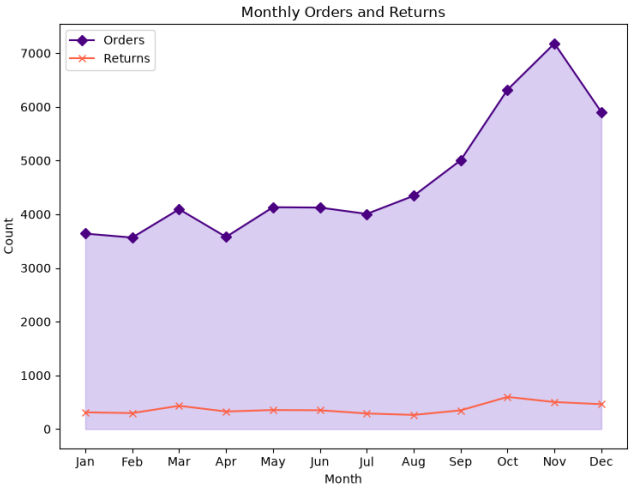
# 6. Customer Rating Distribution (Histogram + KDE)
sns.histplot(orders['Customer Rating'], bins=16, kde=True,
          color='teal', ax=axes[2, 1])
```

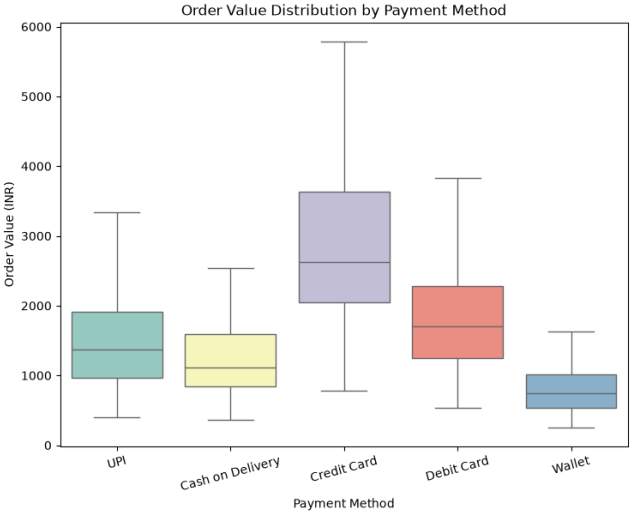
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mean_rating = orders['Customer Rating'].mean()
axes[2, 1].axvline(mean_rating, color='crimson', linestyle='--',
            label=f'Mean = {mean_rating:.2f}')
axes[2, 1].set_title('Customer Rating Distribution')
axes[2, 1].set_xlabel('Rating (1-5)')
axes[2, 1].legend()

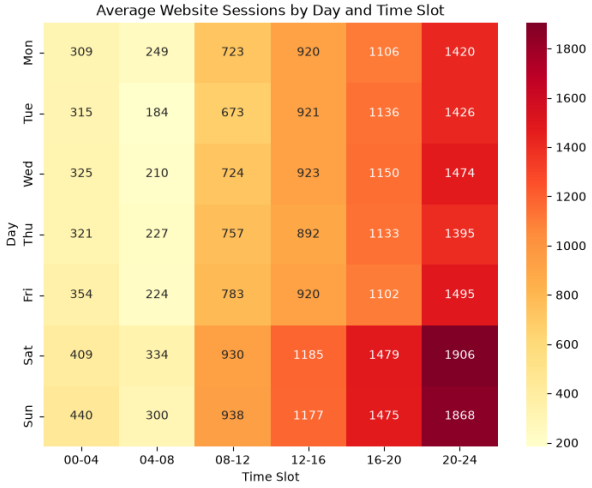
plt.tight_layout()
plt.savefig('ECommerce_Analytics_Dashboard.png')
print("Visualizations successfully saved to ECommerce_Analytics_Dashboard.png")
```

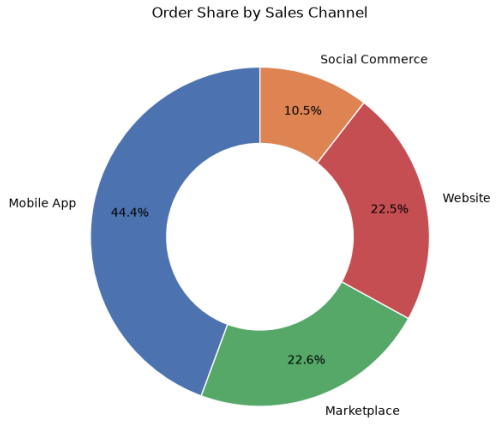
Output:

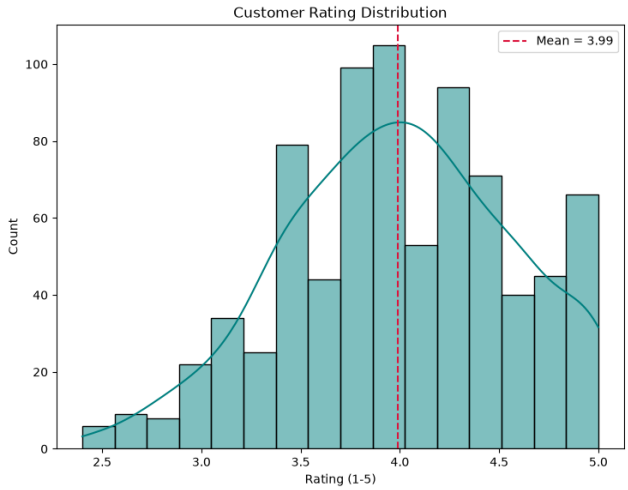

Category Revenue by Region
Category
Electronics
Home & Kitchen
Grocery
Fashion
Beauty
Revenue (INR Lakhs)
0
100
200
300
400
500
600
North
South
East
West
Central
Region


Monthly Orders and Returns
Orders
Returns
Count
0
1000
2000
3000
4000
5000
6000
7000
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Month


Order Value Distribution by Payment Method
Order Value (INR)
0
1000
2000
3000
4000
5000
6000
UPI
Cash on Delivery
Credit Card
Debit Card
Wallet
Payment Method


Average Website Sessions by Day and Time Slot
Day
Mon 309 249 723 920 1106 1420
Tue 315 184 673 921 1136 1426
Wed 325 210 724 923 1150 1474
Thu 321 227 757 892 1133 1395
Fri 354 224 783 920 1102 1495
Sat 409 334 930 1185 1479 1906
Sun 440 300 938 1177 1475 1868
00-04
04-08
08-12
12-16
16-20
20-24
Time Slot
200
400
600
800
1000
1200
1400
1600
1800


Order Share by Sales Channel
Social Commerce
10.5%
Website
22.5%
Marketplace
22.6%
Mobile App
44.4%


Customer Rating Distribution
Mean = 3.99
Count
0
20
40
60
80
100
2.5
3.0
3.5
4.0
4.5
5.0
Rating (1-5)